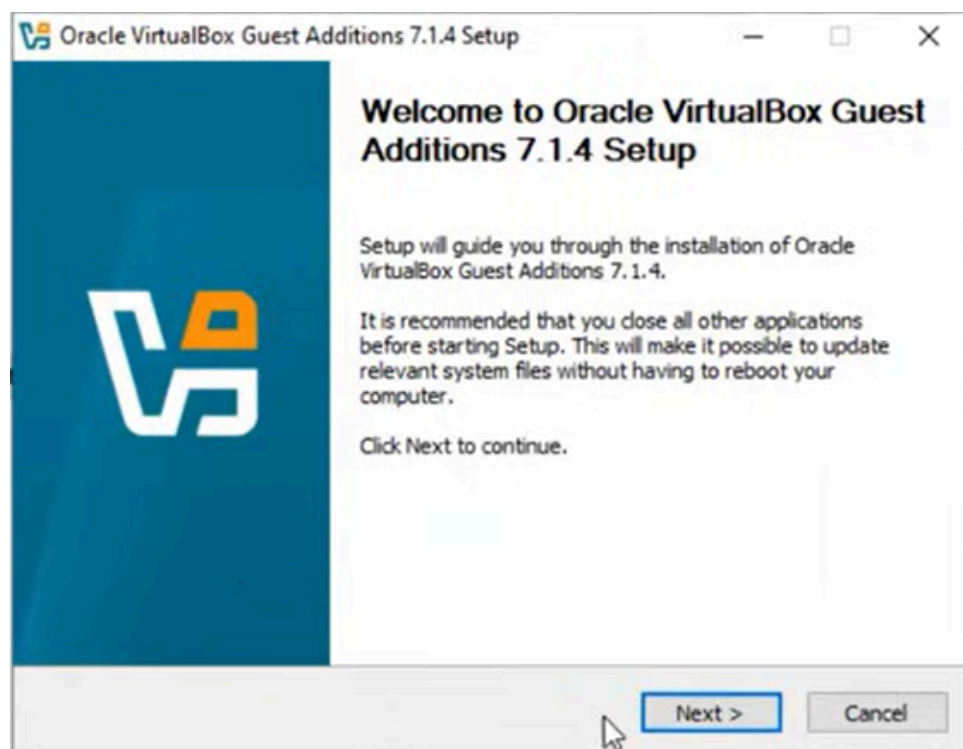
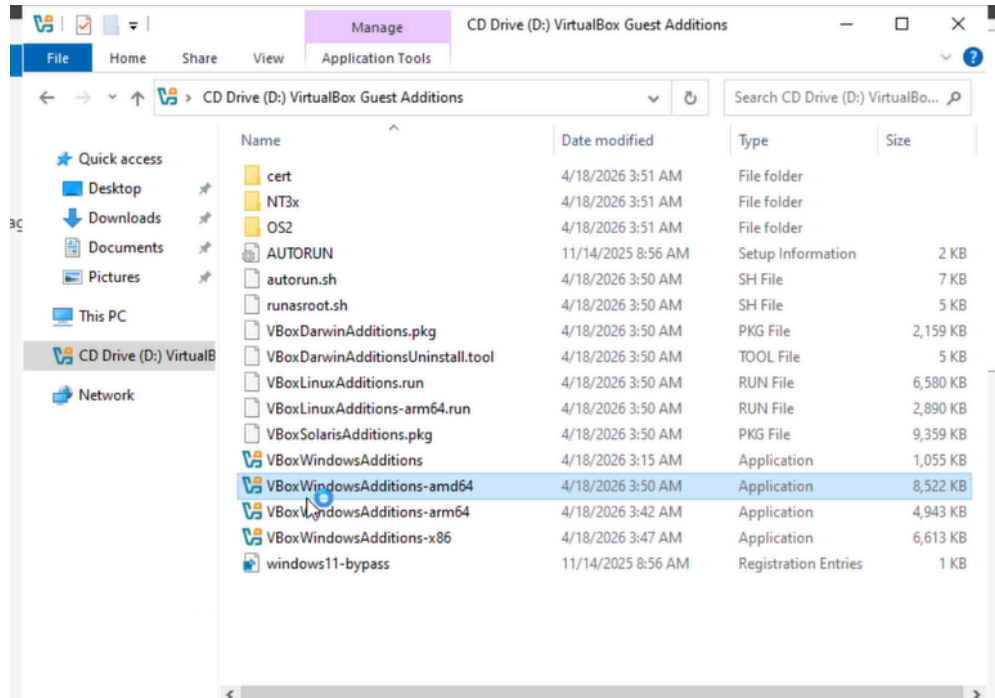


VirtualBox Guest Additions shared folders allow seamless file sharing between your host computer (physical machine) and the guest virtual machine (VM). They map a specific directory on the host to the guest OS, making data transfer easy without needing network configuration.

Add a guest addition device CD

Installing VBoxWindowsAdditions-amd64

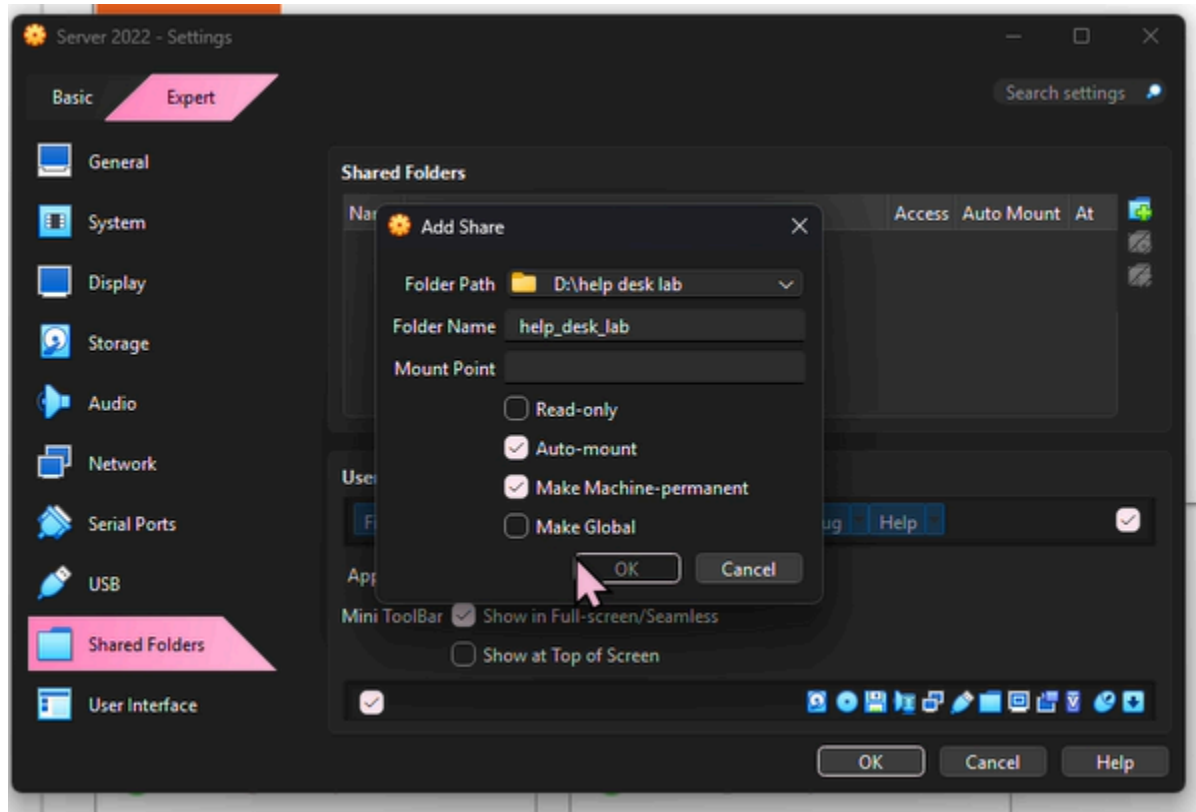


Next trying folder sharing

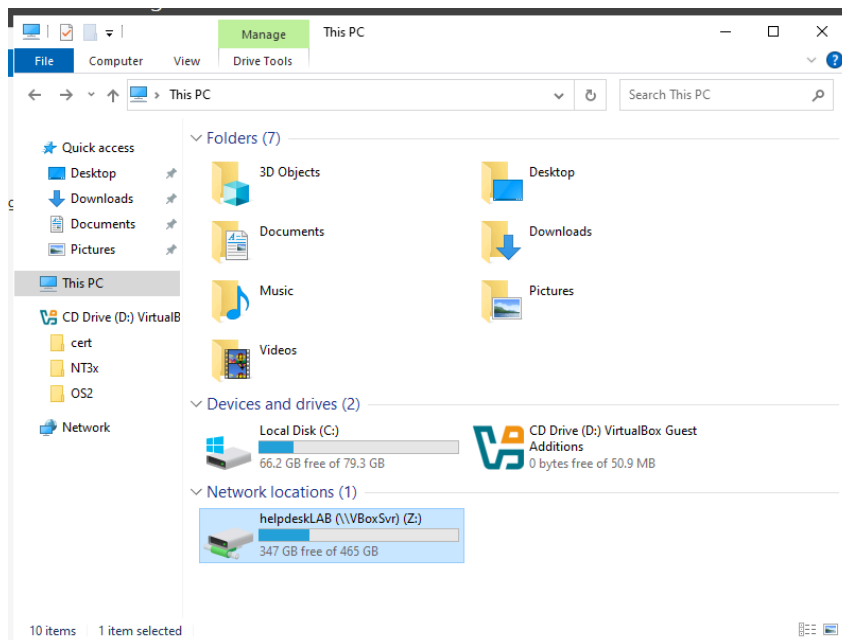
Devices – shared folders settings

Add new folder select other — name and add a folder that you want to share with “helpdesk lab”

Auto-mount make permanent check



Testing real-time sharing

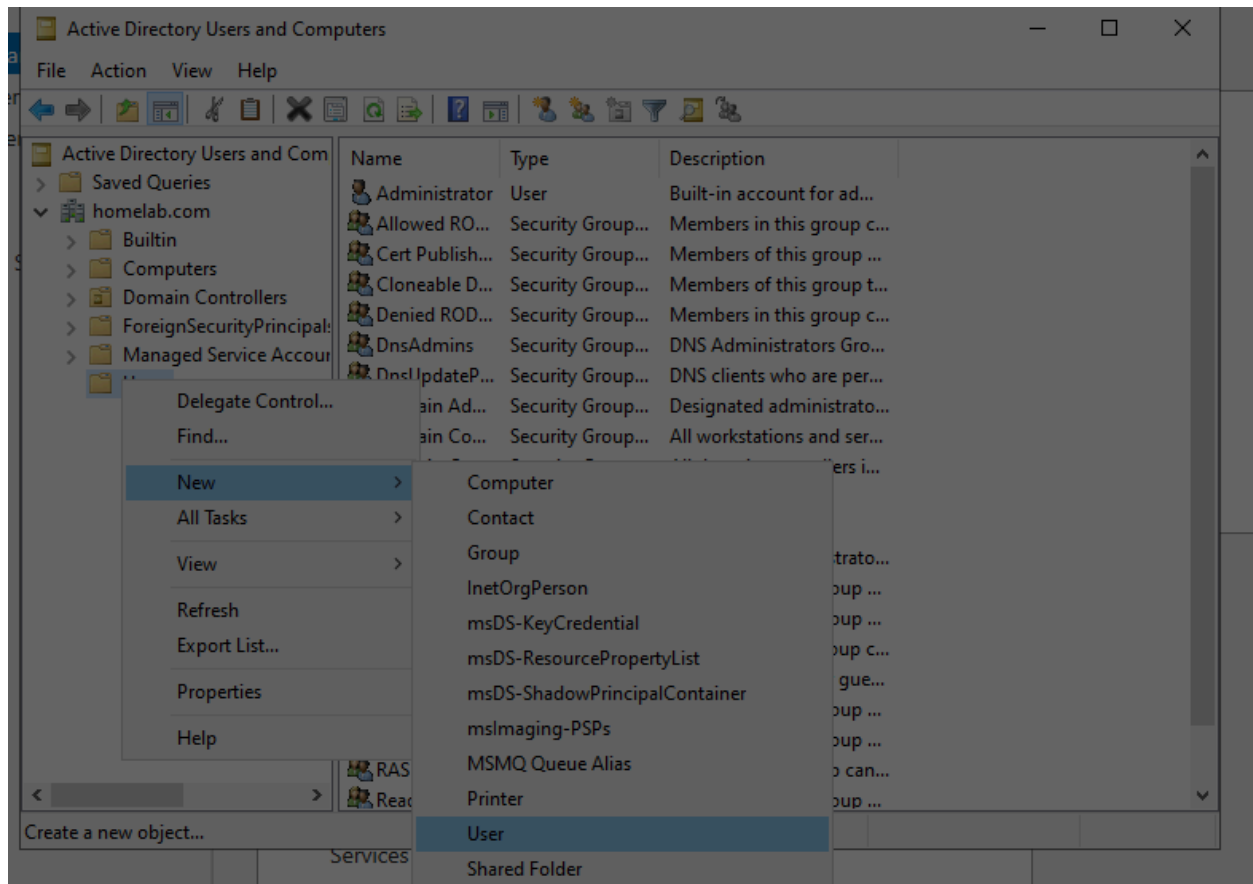


Sharing a document from your original PC will also be shared under this drive

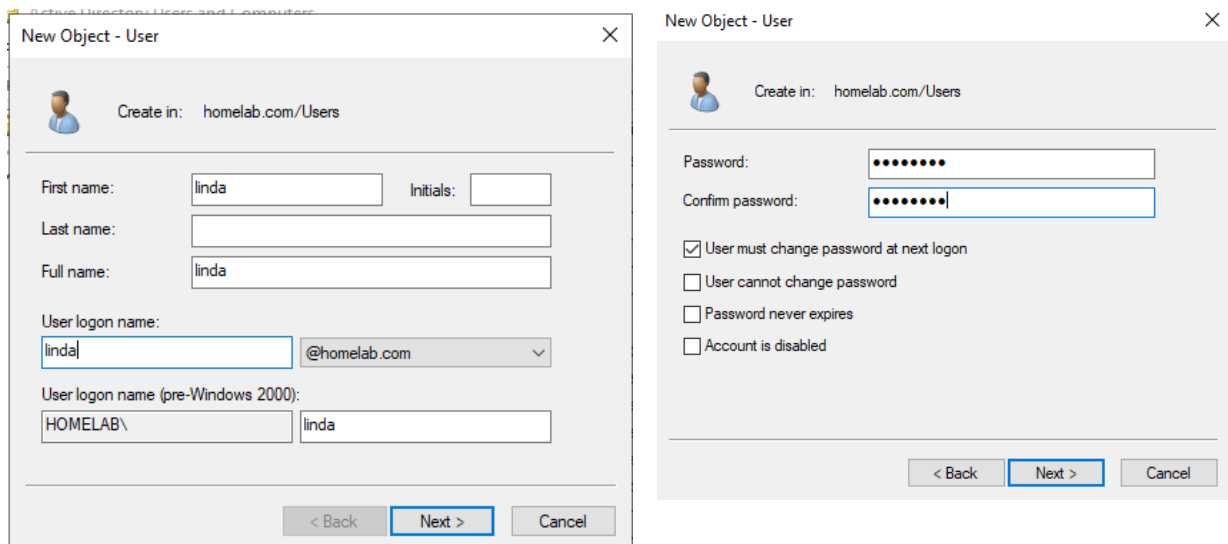
Creating an account by adding new users in active directory

Under tools select active directory users and computers

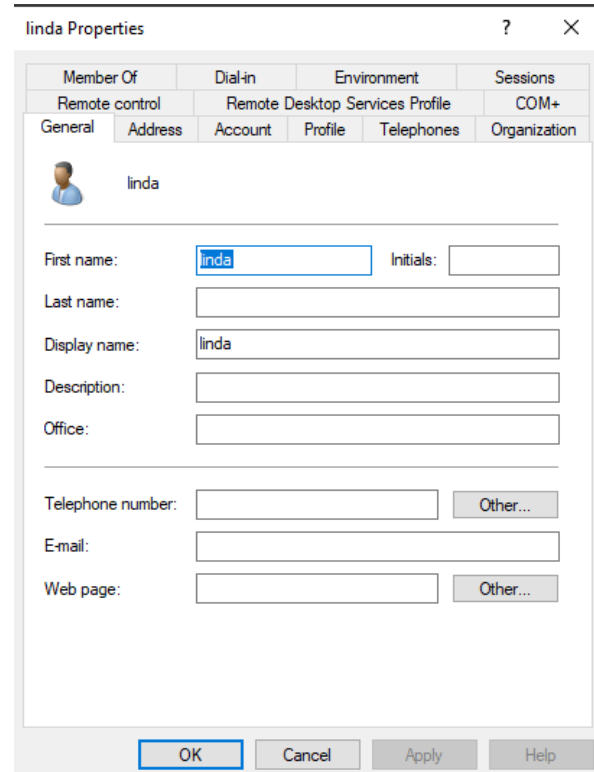
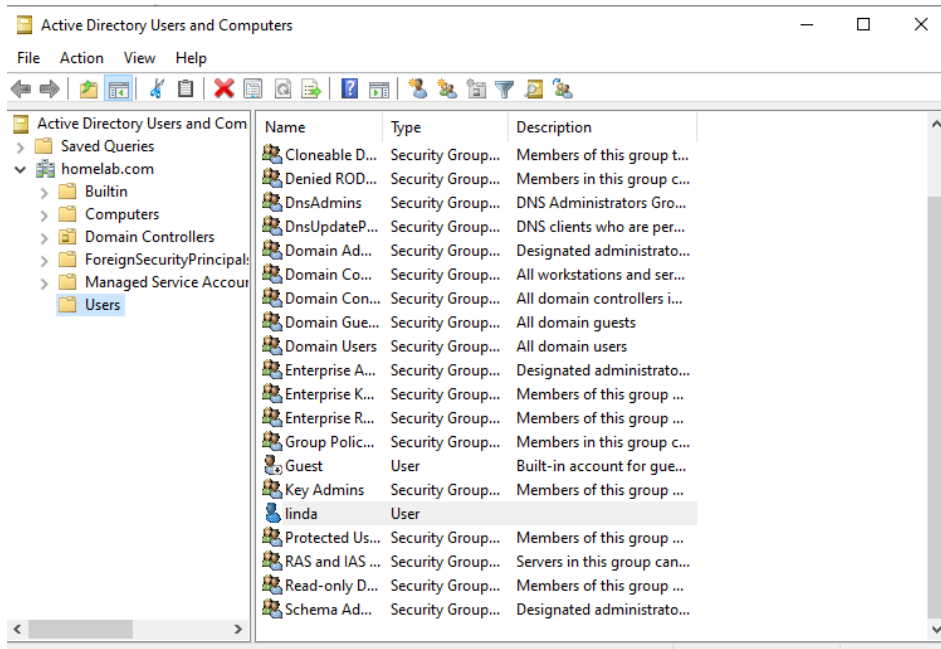
Then expand the domain "homelab.com" – right click on "users" — new — user



Input information of the new account/ review password setting under



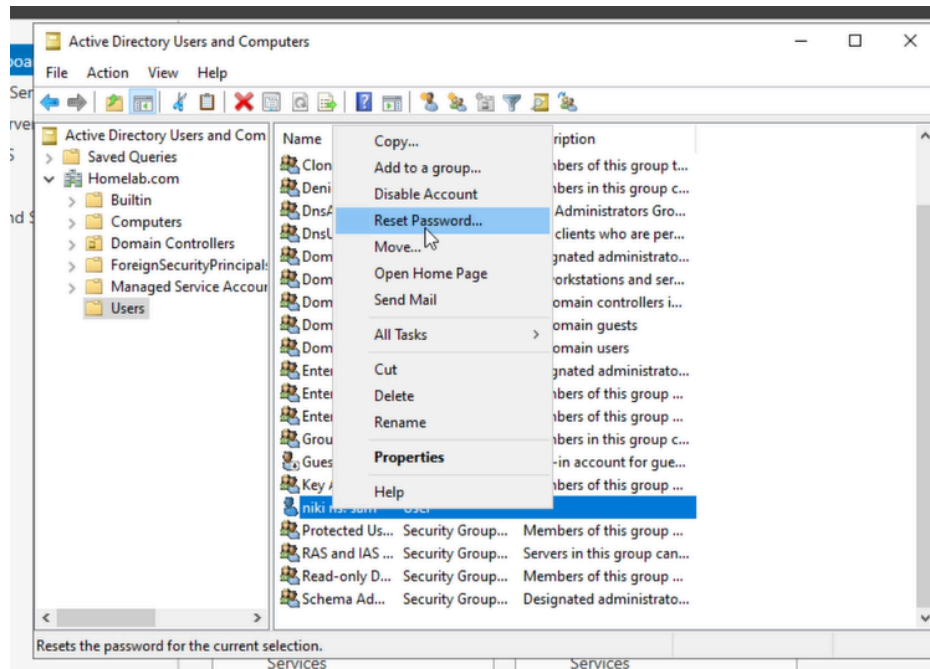
The new user is added and can be clicked on to change or modify properties



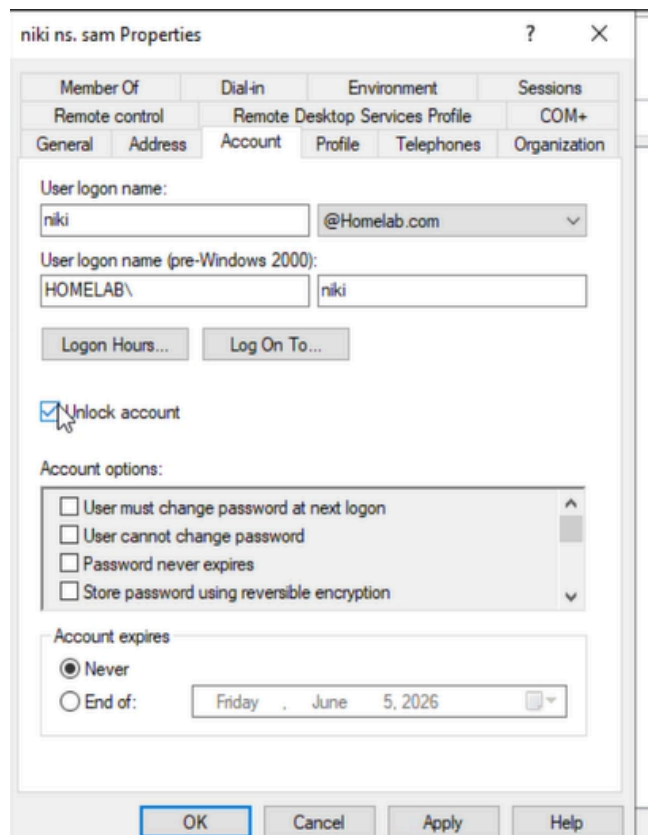
FOR resetting password right click on the user name and options are
Disabling an account in Active Directory (AD) immediately blocks that user or computer from authenticating, preventing access to company resources while keeping the account object intact for future use or auditing.

Can be be enabled back

Resetting a password in Active Directory (AD) immediately changes a user's password to a new value set by an administrator, overriding the old password without needing to know it



Unlocking an account after being locked



to manage or view user accounts within an Active Directory domain, rather than the local machine.

Go on CMD and type the command

Net user (name) /domain

```
C:\Users\Administrator>net user mark /domain
User name                mark
Full Name                mark
Comment
User's comment
Country/region code      000 (System Default)
Account active            Yes
Account expires           Never

Password last set        5/7/2026 11:18:18 AM
Password expires         6/18/2026 11:18:18 AM
Password changeable      5/8/2026 11:18:18 AM
Password required        Yes
User may change password Yes

Workstations allowed     All
Logon script
User profile
Home directory
Last logon               Never

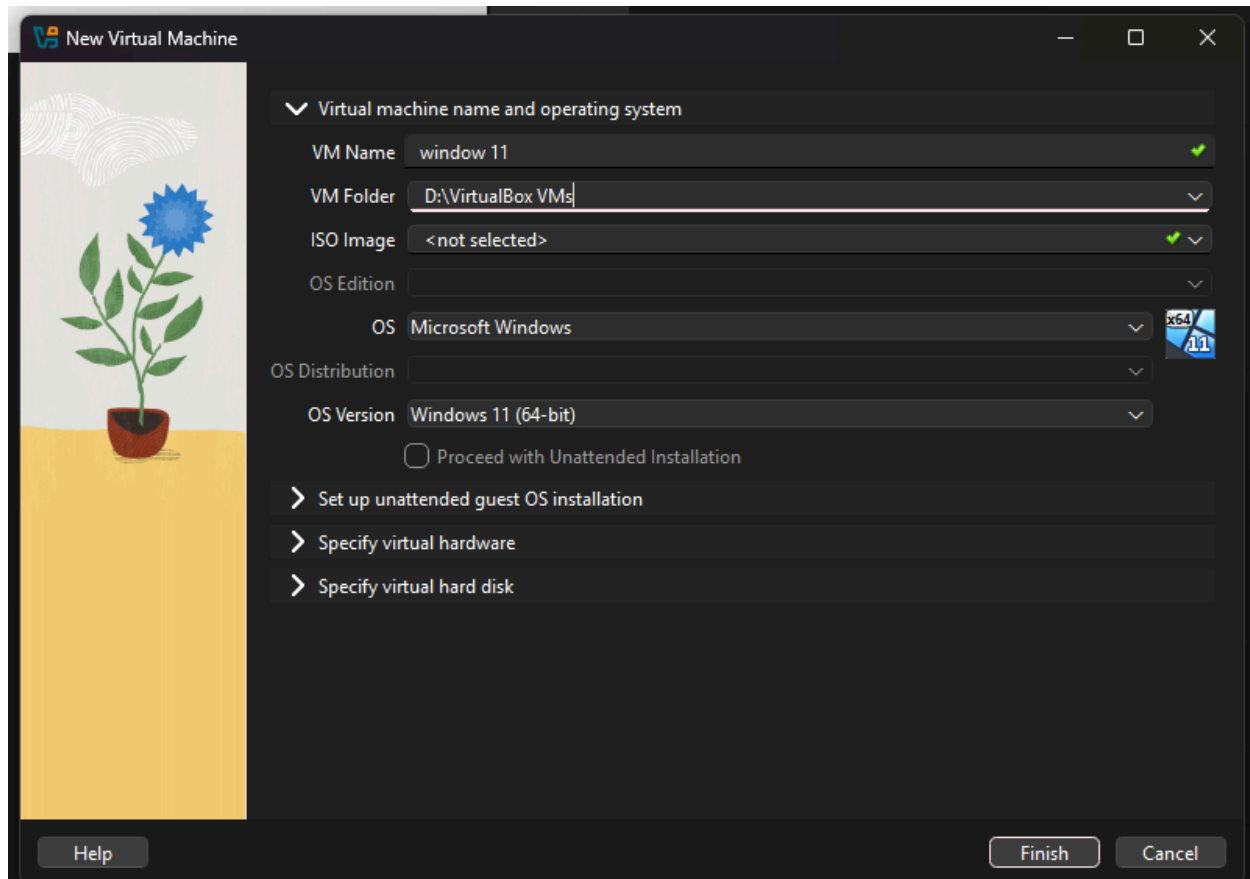
Logon hours allowed      All

Local Group Memberships
Global Group memberships *Domain Users
The command completed successfully.

C:\Users\Administrator>
```

WINDOW 11 INSTALATION

Add a new virtual machine and name choose folder location.



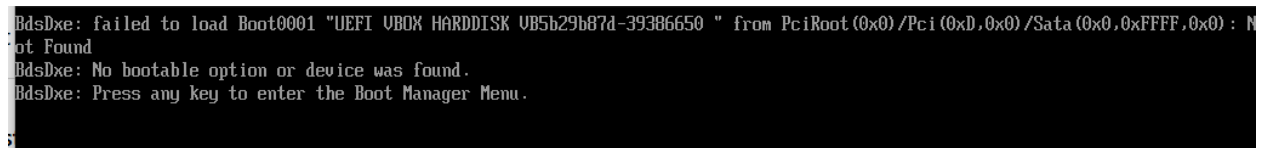
Running the machine will take you to add a window 11 ISO

To do so make sure you have already downloaded

Go to <https://www.microsoft.com/en-us/software-download/windows11>

And download create windows 11 installation media.

If you encountered this problem



shut the machine off completely

Go to the machine setting – storage— click on “empty” and click on the CD icon–

Another problem you might encounter,

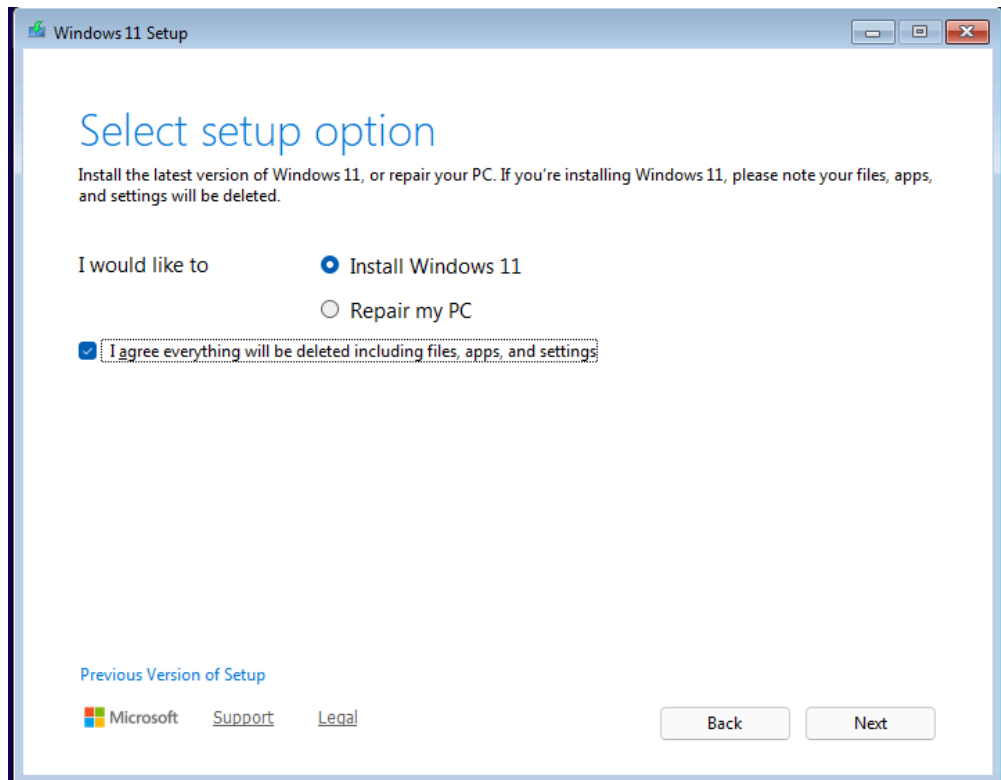
the VM is trying to boot using UEFI but cannot properly find the ISO/boot media.

```
BdsDxe: failed to load Boot0001 "UEFI UBOX CD-ROM UB1-1a2b3c4d " from PciRoot(0x0)/Pci(0xD,0x0)/Sata(0x1,0xFFFF,0x0) : Not Found
```

HOW to fix: go on the machine setting— system — motherboard — uncheck UEFI— ok

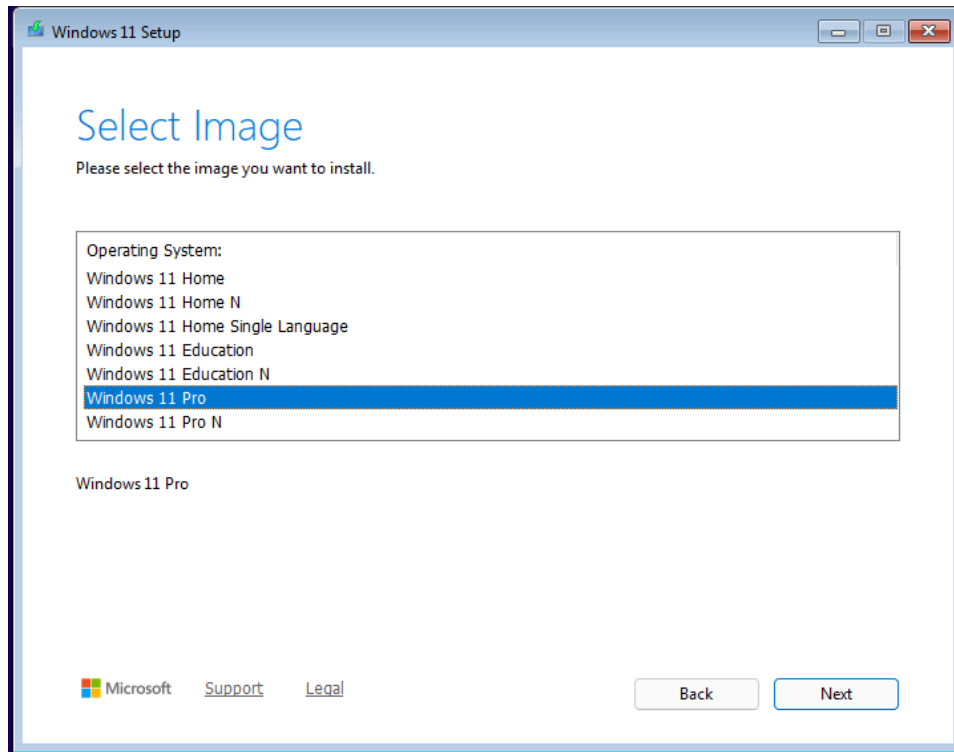
Then go back and check UEFI again — press ok
select window.iso

Keep setting up window 11



Skipping the product key “I don’t have a product key”

Since this is a lab you would want to choose “window 11 pro”



**After window 11 setup to connect to the internet
Go to devices on the top – network— network setting and allow VMs on
promiscuous mode make sure attached to is NAT.
Attached to options:**

NAT

**VM shares your internet.
Best beginner option.**

Bridged

VM acts like a real device on your home network.

**More advanced.
Can be useful later.**

Internal Network

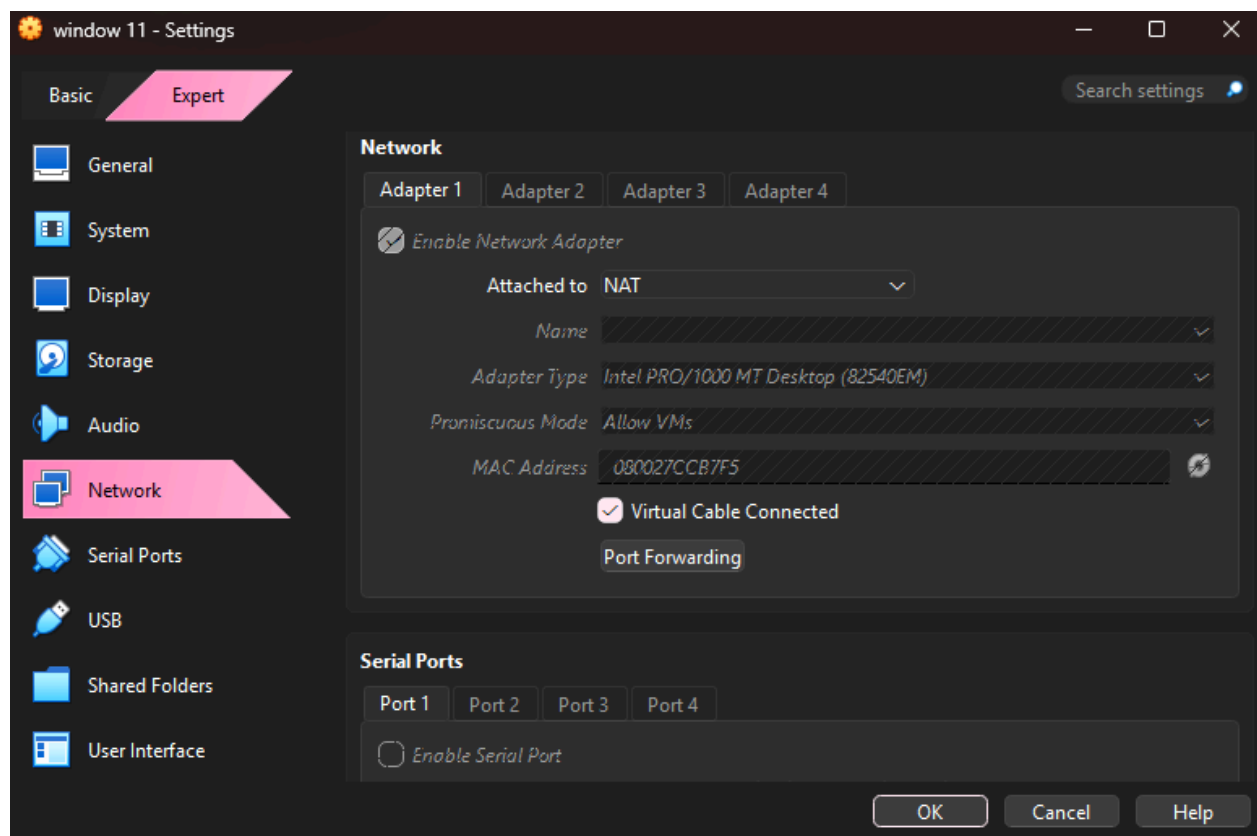
VMs talk ONLY to each other.

Used heavily in AD labs.

Host-Only

VMs can talk to your computer but not the internet.

Useful later too.



Joining Machine to domain (joining employee window 11 to company server 2022)

This can be tricky when selecting a network adapter.

Before running any machine lets change network setting adapter

Server 2022

Adapter 1

Attached to NAT (VGives internet access)

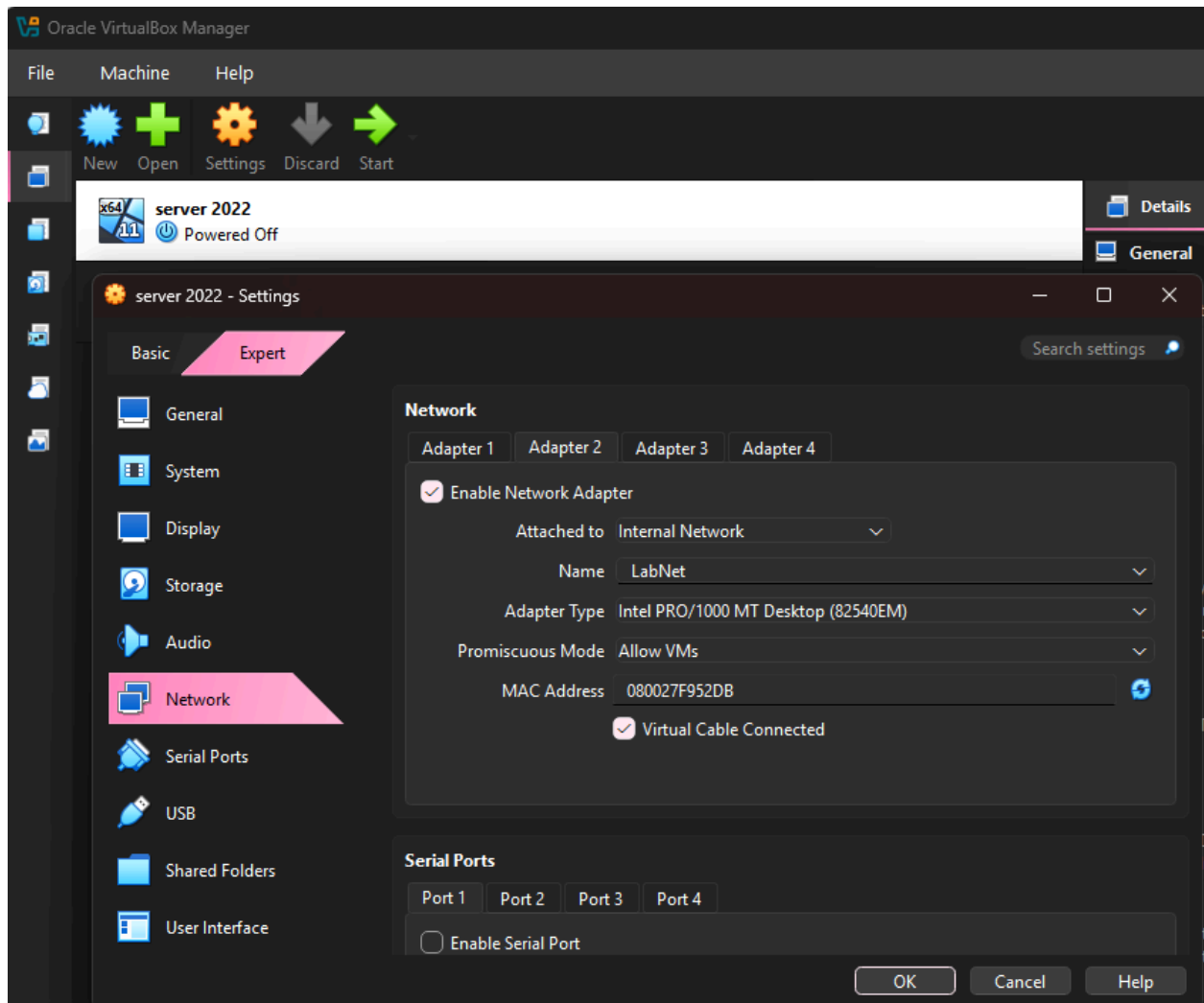
Adapter 2

Check enable network adapter (Gives internet access)

Attached to Internal network (Lets the VMs talk to each other privately)

Name : LabNet (must match on both machines)

Promiscuous mode: allow VM



Window 11

Adapter 1

Attached to NAT (VGives internet access)

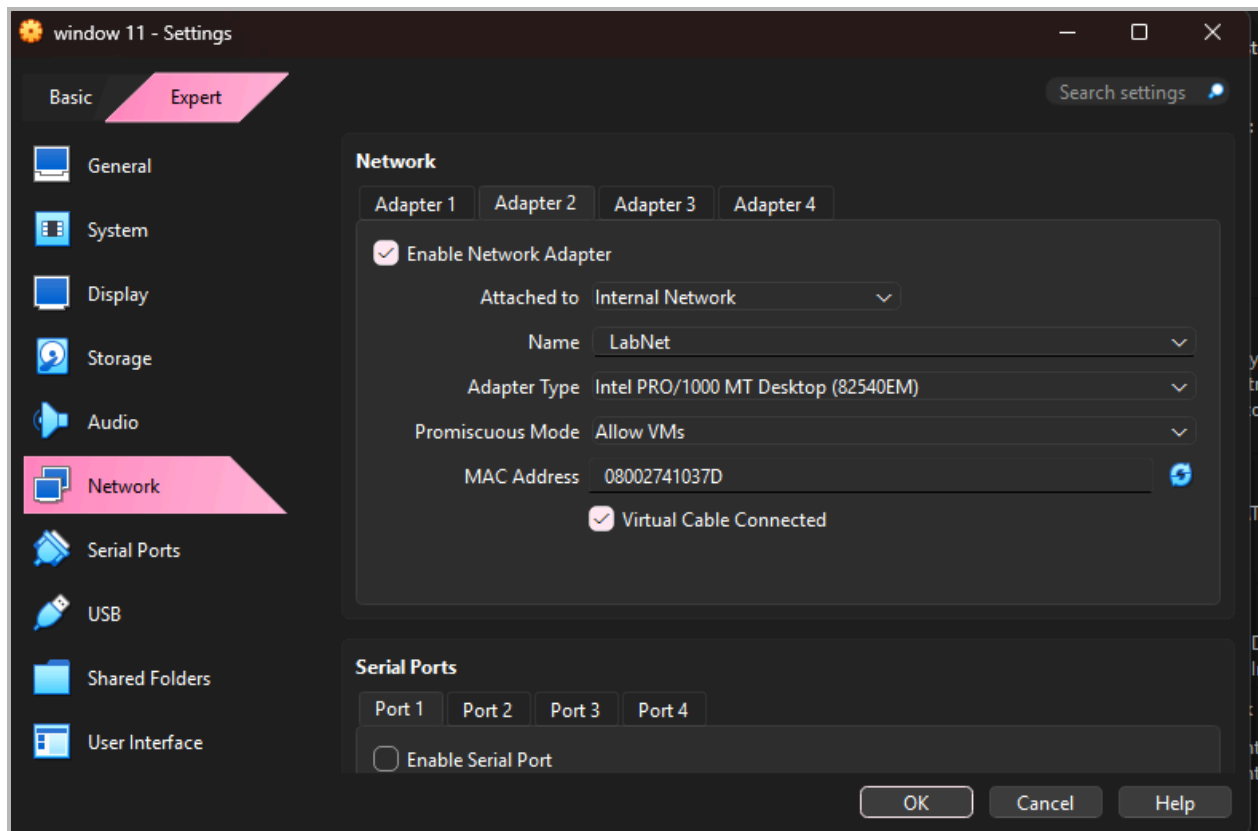
Adapter 2

Check enable network adapter (Gives internet access)

Attached to Internal network (Lets the VMs talk to each other privately)

Name : LabNet (must match on both machines)

Promiscuous mode: allow VM



Troubleshooting pinging fail

If you try to ping the domain it would fail

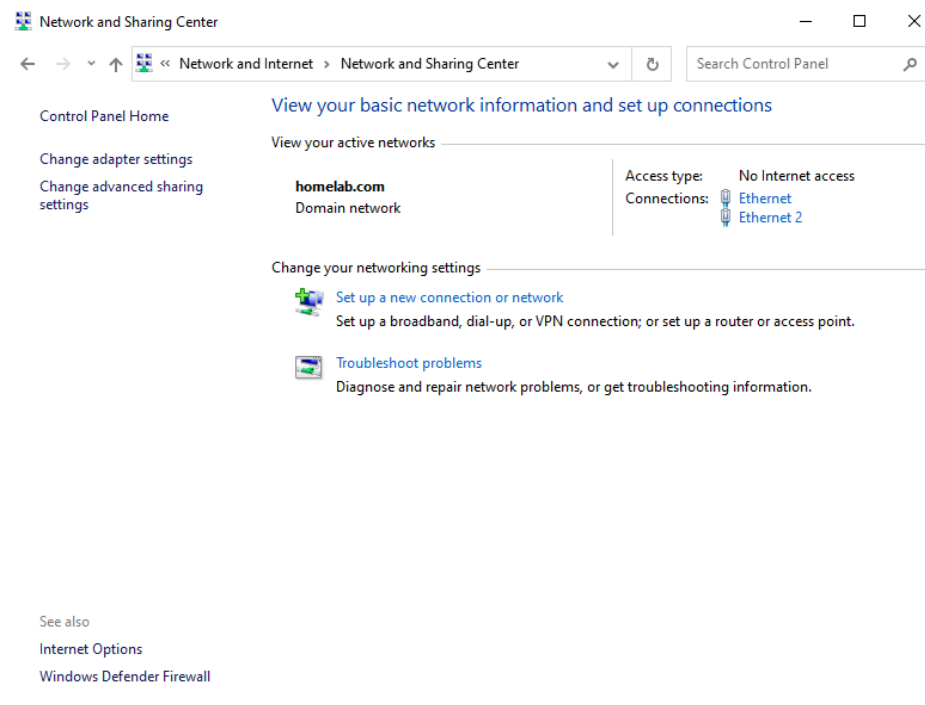
Must set static IP addresses, subnet mask, and default gateway to have both machines ping each other for both ethernet adapter 1 and adapter 2

STEPS:

One both machines

1. Control panel— network and internet— view network status and tasks
2. Click ethernet— properties— internet protocol version 4 (TCP/IPV4)
3. Set the ip address and DNS server as shown in the image below. One difference between to machines server 2022: ip address (10.1.10.3) window 11: ip address (10.1.10.2)
4. Click ethernet 2 —Click ethernet— properties— internet protocol version 4 (TCP/IPV4)

Server 2022: ip address 192.168.10.10 DNS server: 192.168.10.10
Window 11:ip address 192.168.10.20 DNS server: 192.168.10.10



Internet Protocol Version 4 (TCP/IPv4) Properties

✕

General

Ethernet 1

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

10 . 1 . 10 . 3

Subnet mask:

255 . 0 . 0 . 0

Default gateway:

10 . 1 . 10 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

10 . 1 . 10 . 2

Alternate DNS server:

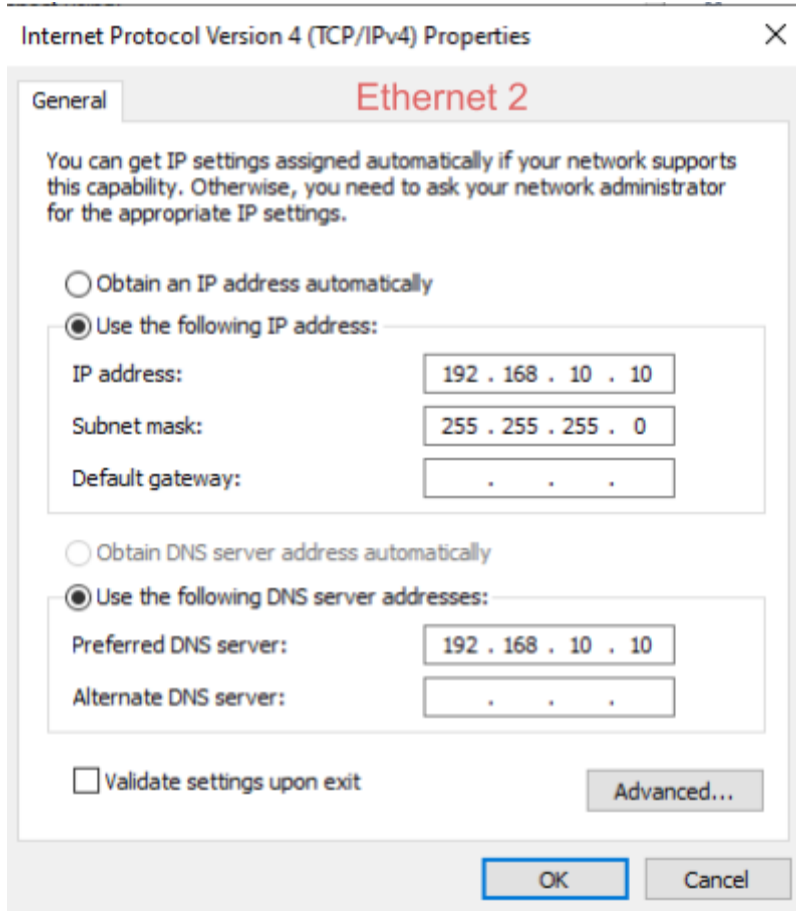
10 . 1 . 10 . 1

☐ Validate settings upon exit

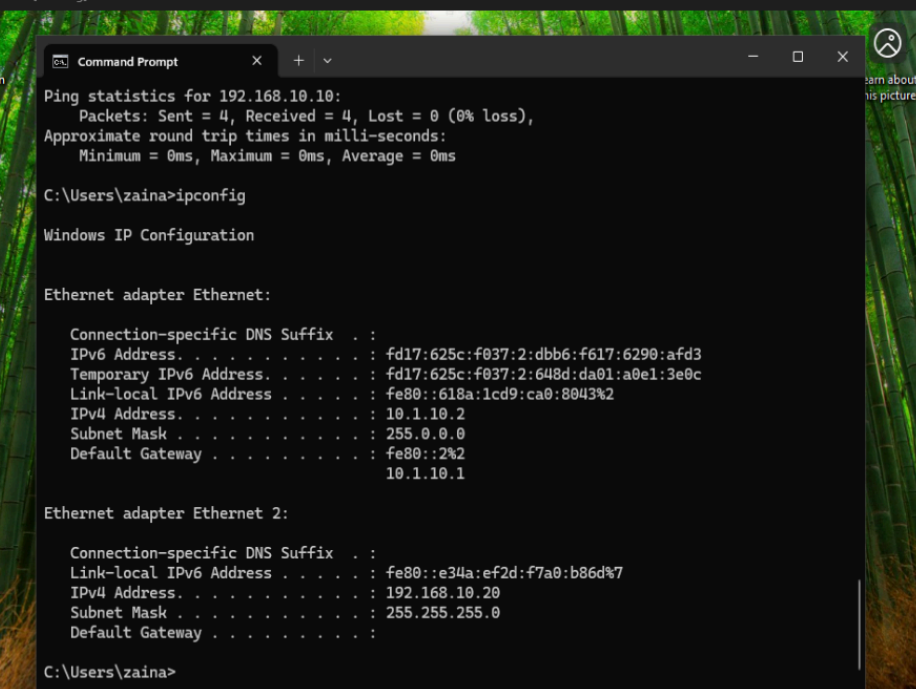
Advanced...

OK

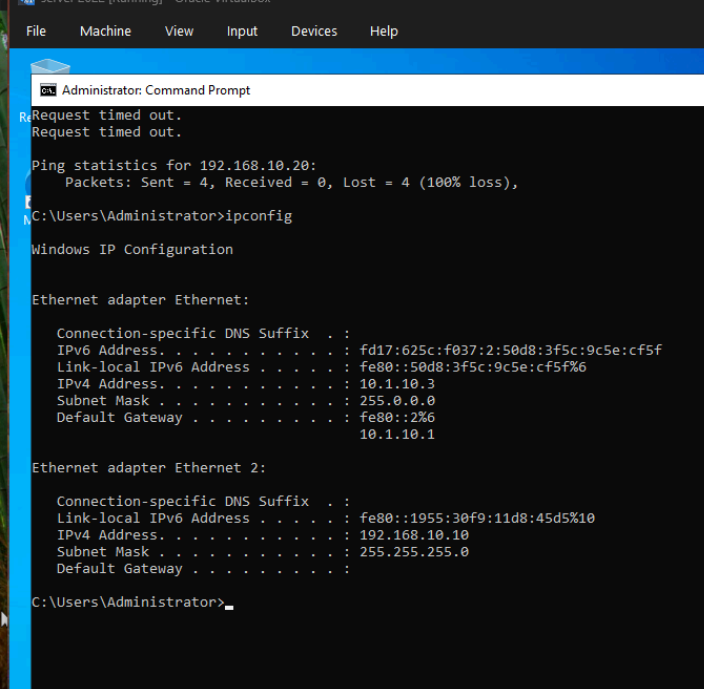
Cancel



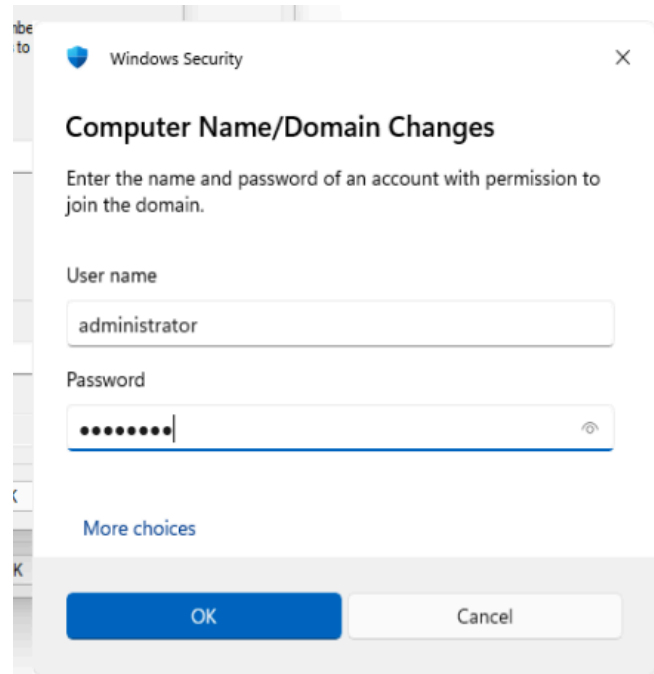
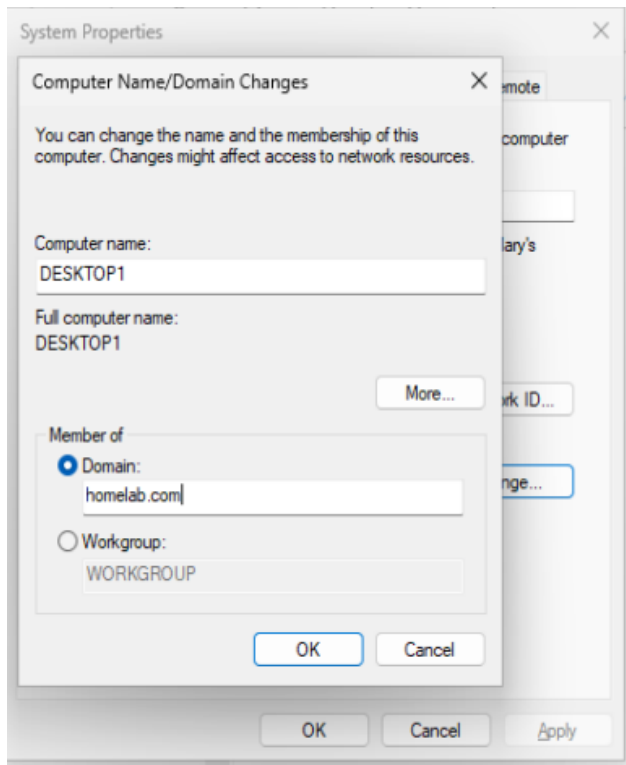
ow 11 [Running] - Oracle VirtualBox



server 2022 [Running] - Oracle VirtualBox



I joined the Windows 11 machine to the domain through the system settings using domain administrator credentials. Once joined successfully, the client machine was able to authenticate domain users created in Active Directory.



Then I was successful in remotely accessing the server from my user account